Project Report: Airbnb Market Dynamics — Chicago vs. New Orleans

Project Category: Business Intelligence & Exploratory Data Analysis

Tools & Technologies: Python (Pandas, NumPy), Power BI Desktop, DAX

Datasets Analyzed: Chicago_listings (1).csv, New Orleans_listings.csv

1. Executive Summary

This project presents an end-to-end comparative market analysis of short-term rental listings across two prominent United States tourism and business hubs: Chicago, Illinois, and New Orleans, Louisiana. Using a dual-stage analytical workflow—data integration and feature engineering in Python followed by business intelligence modeling in Power BI—the study evaluates 14,021 active listings to uncover structural supply patterns, pricing mechanisms, host operations, and listing availability.

The analysis confirms that the combined market represents an average nightly rate of $294.40 with an average annual availability of 240.76 days. Chicago commands a 54.21% market share (7,601 listings), while New Orleans comprises 45.79% (6,420 listings). Both markets exhibit heavy inventory concentration in "Entire Home/Apartment" units (10,926 total properties) and are structurally dominated by multi-unit commercial operators holding three or more listings. These empirical findings provide actionable benchmarks for hospitality investors, property management firms, and market researchers seeking data-driven positioning strategies.

2. Business Problem & Project Objectives

2.1 Problem Statement

Short-term rental markets vary substantially by municipal regulations, tourism seasonality, and demographic demand drivers. Real estate operators and prospective investors entering unfamiliar metropolitan territories face operational risks if they rely solely on high-level national metrics. To make viable capital deployment decisions, stakeholders require cross-city comparative visibility into:

- How listing density and room inventories differ across primary urban districts.
- The true baseline nightly rates across different accommodation classes.
- Whether markets are sustained by grassroots home sharers or saturated by institutional property managers.

- How listing availability interacts with minimum booking requirements.

2.2 Core Objectives

- **Data Pipeline Execution:** Ingest, clean, reconcile, and merge separate city-level Airbnb datasets into an integrated tabular schema using Python and Pandas.
- **Feature Engineering:** Design categorical segmentations for nightly pricing, host portfolio concentration, annual calendar availability, and guest review engagement.
- **Cross-Market Comparison:** Quantify market volume shares, geographic inventory density, and room-type distributions between Chicago and New Orleans.
- **Interactive Dashboard Development:** Construct a 4-page Power BI dashboard (`Overview`, `Property`, `Pricing`, `Host`) to enable multi-layered business exploration.
- **Strategic Recommendations:** Deliver evidence-grounded operational guidance for pricing positioning, portfolio scaling, and regulatory awareness.

3. Dataset Architecture & Cleaning Methodology

3.1 Source Data Ingestion & Reconciliation

- The raw data inputs consisted of two independent CSV files containing municipal listing snapshots:
- Chicago_listings (1).csv (Chicago market records)
- New Orleans_listings.csv (New Orleans market records)
- Each dataset was tagged with an explicit identifier column (df['city'] = 'Chicago' and df['city'] = 'New Orleans') prior to vertical unification via pd.concat().

3.2 Cleaning Operations & Imputation Logic

To establish analytical consistency, the raw data underwent targeted preprocessing steps:

Python

```
import numpy as np
import pandas as pd

# 1. Deduplication and critical null removal
df = df.dropna(subset=['price', 'latitude', 'longitude', 'host_id'])
df = df.drop_duplicates(subset=['id'])

# 2. Structural column management
if 'neighbourhood_group' in df.columns and
df['neighbourhood_group'].isnull().all():
  df = df.drop(columns=['neighbourhood_group'])

# 3. Categorical and text imputation
df['reviews_per_month'] = df['reviews_per_month'].fillna(0)
df['last_review'] = df['last_review'].fillna('No Reviews')
df['host_name'] = df['host_name'].fillna('Unknown Host')
df['license'] = df['license'].fillna('Unlicensed/Exempt')

# 4. Text standardization
df['neighbourhood'] =
df['neighbourhood'].astype(str).str.title().str.strip()
df['room_type'] = df['room_type'].astype(str).str.title().str.strip()
```

- **Handling Missing Coordinates & Identifiers:** Rows missing critical spatial attributes (`latitude`, `longitude`), financial indicators (`price`), or owner identifiers (`host_id`) were purged, as geospatial and financial attribution cannot be reliably imputed.
- **Review Metric Assumptions:** Null values in `reviews_per_month` were imputed with `0`, reflecting listings that have recorded zero completed guest stays or review submissions.
- **License Field Semantics:** Missing values in `license` were designated as `Unlicensed/Exempt`. In municipal short-term rental data, an unpopulated license field does not automatically imply non-compliance; it frequently reflects pending verifications, legacy exemptions, or hotel accommodations operating under distinct commercial lodging statutes.

4. Feature Engineering Framework

To translate granular continuous variables into business-friendly categorical dimensions, four dedicated features were engineered:

4.1 Price Tier Classification (price_tier)

Segmented nightly rates into intuitive operational tiers:

- **Budget:** < $75
- **Standard:** $75 – $200
- **Premium:** $201 – $500
- **Luxury:** > $500

4.2 Host Portfolio Concentration (host_type)

Constructed by calculating total properties per owner across the combined dataset:

$$\text{total_host_listings} = \text{Count of listing IDs grouped by host_id}$$

- **Casual Host:** 1 to 2 listings (grassroots, single-unit, or individual home sharers).
- **Commercial Host:** 3 or more listings (institutional property management entities, co-hosts, or syndicates).

4.3 Availability Tiers (availability_tier)

Segmented listings based on annual active calendar days (availability_365):

- **Fully Booked / Inactive:** 0 days
- **Low Availability:** 1 – 89 days
- **Medium Availability:** 90 – 269 days
- **High Availability:** 270 – 365 days

4.4 Guest Engagement Tiers (engagement_tier)

Categorized historical stay activity proxy using cumulative guest feedback (number_of_reviews):

- **No Reviews:** 0 reviews
- **Low Engagement:** 1 – 10 reviews
- **Moderate Engagement:** 11 – 50 reviews
- **High Engagement:** > 50 reviews

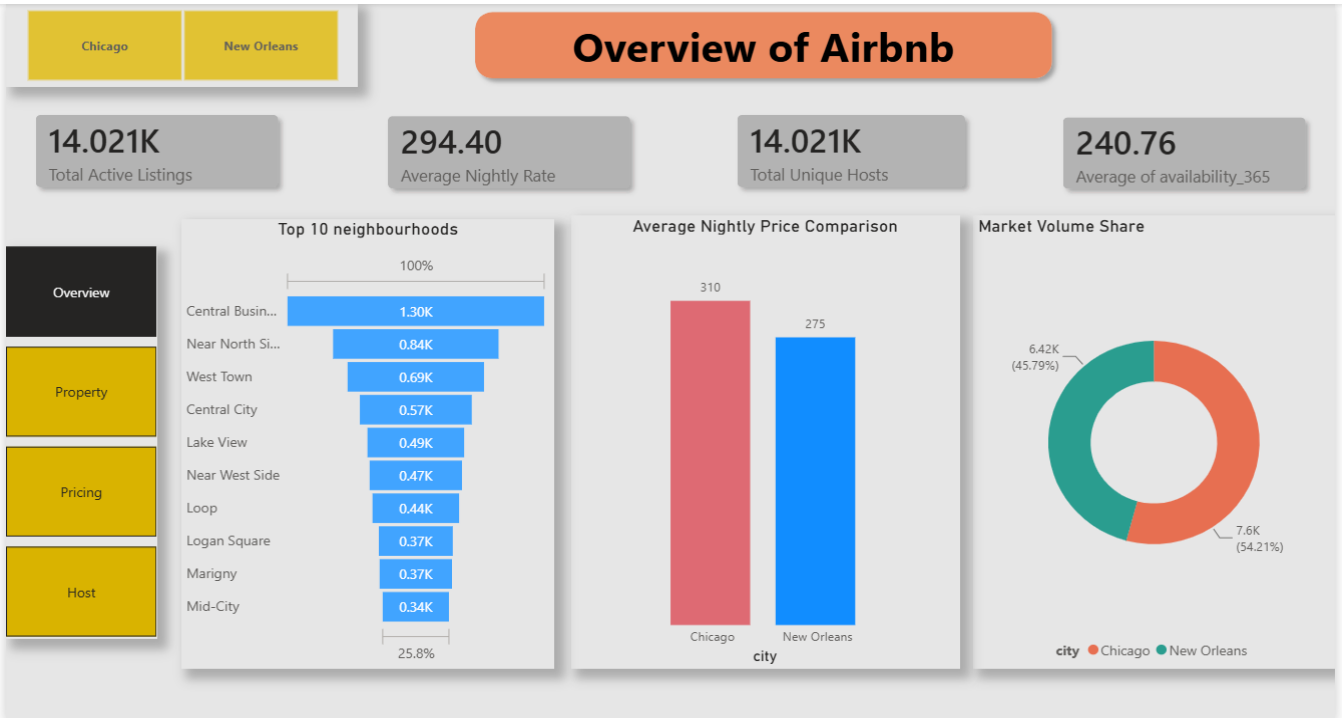
5. Power BI Dashboard Breakdown & Empirical Findings

The cleaned data (Cleaned_Airbnb_Chicago_NewOrleans.csv) powers an interactive 4-page Power BI report structured for executive exploration.

Page 1: Overview

The opening view establishes overall scale and market share benchmarks between the two cities.

- **Executive KPI Cards:**
  - **Total Active Listings:** 14,021 properties
  - **Average Nightly Rate:** $294.40
  - **Total Unique Hosts:** 14,021 host records tracked
- **Average Availability:** 240.76 days available per year
- **Market Volume Share (Donut Chart):** Chicago holds 54.21% of the total inventory (7.6K listings), while New Orleans represents 45.79% (6.42K listings).
- **Average Nightly Price Comparison (Column Chart):** Chicago commands a higher average nightly rate at $310 per night compared to New Orleans at $275 per night.
- **Top 10 Neighbourhoods (Funnel Visual):** Listing volume is heavily concentrated in commercial and tourist corridors. The top neighborhoods across the markets include



Central Business District (1.30K listings), Near North Side (0.84K), West Town (0.69K), Central City (0.57K), and Lake View (0.49K).

Page 2: Property Analysis

This section analyzes property type composition, booking constraints, and calendar availability.

- **Room Type Share Composition & Inventory Matrix:**
  - **Entire Home/Apt:** 10,926 units (5,813 in Chicago; 5,113 in New Orleans).
  - **Private Room:** 2,813 units (1,693 in Chicago; 1,120 in New Orleans).
  - **Hotel Room:** 248 units (65 in Chicago; 183 in New Orleans).
  - **Shared Room:** 34 units (30 in Chicago; 4 in New Orleans).
- **Annual Listing Availability Tiers (Column Chart):** A substantial majority of properties fall into the "High Availability (> 270 days)" bracket (approx. 6.9K listings), followed by "Medium Availability" (approx. 5.3K listings), and "Low Availability (< 90 days)" (approx. 1.8K listings). Only a minimal fraction are classified as Fully Booked/Inactive (0.0K).
- **Minimum Stay Requirement Trends (Bar Chart):** Minimum stay requirements vary by property type. Shared Rooms carry the highest average minimum stay (approx. 17 nights), followed by Entire Homes/Apartments (approx. 14 nights) and Private Rooms (approx. 13 nights). Hotel Rooms mandate the lowest minimum
- commitment (approx. 3 nights).



Page 3: Pricing Analysis

This section evaluates rate structures, premium inventory, and demand elasticity.

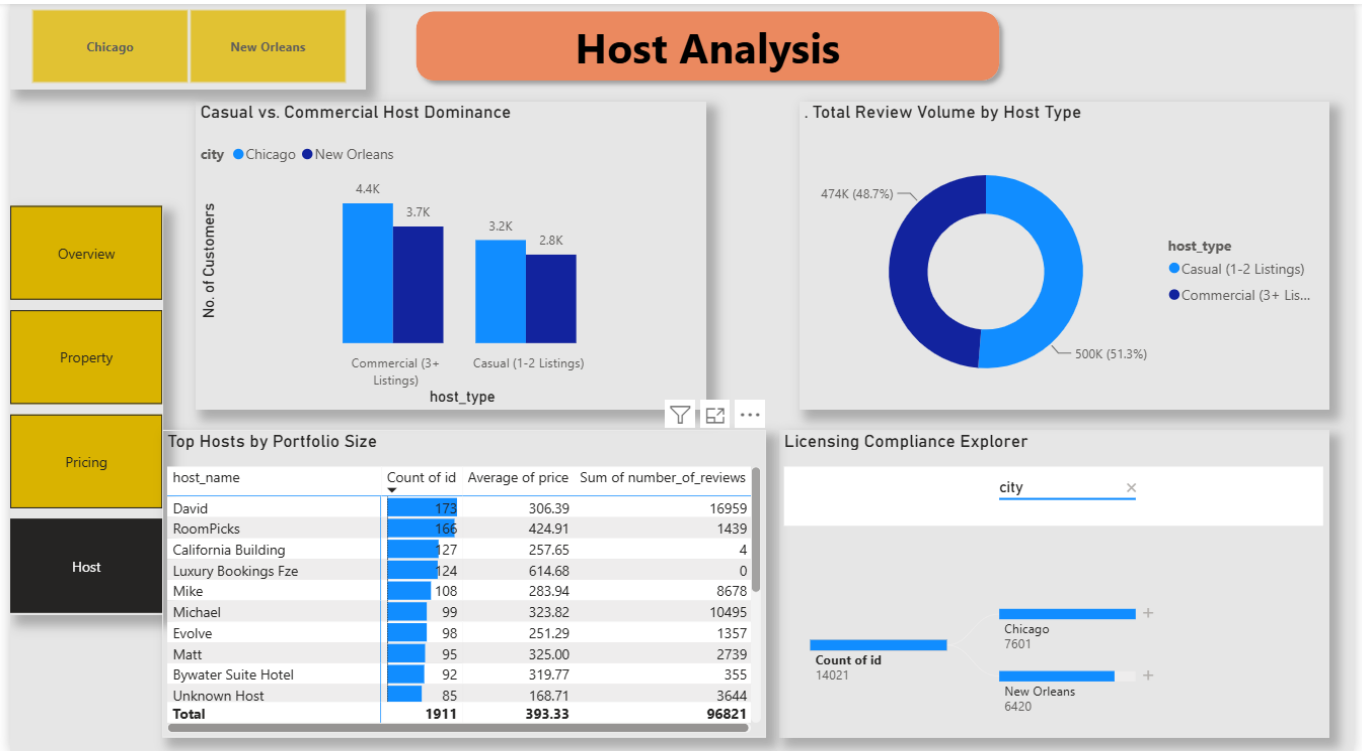
- **Price Tier Breakdown (Clustered Bar Chart):** In both cities, the largest share of listings falls within the **Premium ($201–$500)** tier (3.2K in Chicago; 2.4K in New Orleans) and the **Standard ($75–$200)** tier (2.5K in Chicago; 2.8K in New Orleans). Luxury listings (> $500) account for 1.1K units in Chicago and 0.6K in New Orleans.
- **Nightly Rate by Room Type:** Hotel Rooms hold the highest average nightly rate, approaching $500 in Chicago, followed by Entire Home/Apt inventory. Shared Rooms exhibit the lowest average rate across both markets.
- **Top 10 Most Expensive Neighbourhoods:** Chicago dominates the highest-rate brackets, led by the Loop, Near North Side, Touro, Lake Catherine, Lincoln Park, and West Town.
- **Price Elasticity vs. Booking Demand (Scatter Plot):** Plotting price against reviews_per_month demonstrates classic demand elasticity: listings priced below $200 record the highest volume of frequent guest reviews, while rates scaling toward luxury levels (> $500) exhibit a tapering review frequency.



This view measures operator concentration, professionalization, and market distribution.

- **Casual vs. Commercial Host Dominance (Bar Chart):** Commercial operators (3+ listings) control the majority of listings in both cities (approx. 4.4K listings in Chicago; 3.7K in New Orleans) compared to Casual hosts (3.2K in Chicago; 2.8K in New Orleans).

- **Total Review Volume by Host Type (Donut Chart):** Casual hosts account for 51.3% of total cumulative guest reviews (500K reviews), while Commercial hosts generate 48.7% (474K reviews).
- **Top Hosts by Portfolio Size (Leaderboard Table):** The commercial landscape is led by prominent operators:
    - *David:* 173 listings (Average price: $306.39; 16,950 total reviews)
    - *RoomPicks:* 166 listings (Average price: $424.91; 1,439 total reviews)
    - *California Building:* 127 listings (Average price: $257.65; 4 total reviews)
    - *Luxury Bookings Pzc:* 124 listings (Average price: $614.08; 0 reviews)
    - *Mike:* 108 listings (Average price: $283.94; 8,678 total reviews)
    - *Michael:* 99 listings (Average price: $323.82; 10,495 total reviews)



6. Business Recommendations

1. **Strategic Inventory Positioning (The $100–$225 Sweet Spot):**
    - While average nightly rates sit between $275 and $310, the demand elasticity scatter plot reveals that listing velocity (measured via reviews per month) concentrates heavily below $200.
    - Operators seeking to optimize annualized occupancy rather than speculative peak rates should price entire home units between $150 and $225 during non-peak cycles.
2. **Capitalize on the Entire Home Preference:**
    - Standalone Entire Home/Apartment configurations represent 77.9% of all market inventory (10,926 of 14,021 listings).
    - Real estate investors should prioritize multi-bedroom residential assets that can host families and group travelers, rather than small single-room shared models, which hold less than 0.3% market share.

3. **Benchmarking Against Institutional Competitors:**
    - Commercial property operators (e.g., David, RoomPicks, California Building) operate large portfolios exceeding 100 listings each.
    - Independent operators cannot compete with institutional managers on volume discounts. Instead, they should focus on hyper-local guest experiences, faster review turnaround, and personalized hospitality—leveraging the finding that casual hosts generate over 51% of total market reviews.
4. **Monitoring Minimum Stay Mandates:**
    - With average minimum stay requirements hovering around 14 nights for entire apartments, hosts must align their booking rules with local zoning laws (e.g., Chicago's Shared Housing Ordinance and New Orleans Short-Term Rental regulations). Implementing 3-to-5-night minimums during weekend events, combined with extended 30-day corporate stay options, mitigates regulatory exposure.

7. Conclusion

This project delivers a data-backed comparison of short-term rental dynamics in Chicago and New Orleans. By coupling automated Python data preprocessing with interactive Power BI dashboards, the analysis confirms that both markets are structurally dominated by entire residential properties and commercial operators, with Chicago maintaining higher top-end listing prices and larger overall market volume. The resulting dashboards and analytical models offer a scalable blueprint for ongoing municipal rental market evaluations.